PREDICTIVE WORKFORCE PLANNING

Using HR Recruitment Analytics

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Dataset: HRDataset_v14.csv | **Source:** Kaggle

Tools: Python | Jupyter Notebook | scikit-learn | pandas | matplotlib

Data Science Internship Project

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1. Executive Summary

This report presents a complete predictive workforce planning analysis using historical HR recruitment data from a fictional organisation. The dataset contains 311 employee records spanning 2006 to 2018 across six departments. The analysis forecasts future hiring needs using Linear Regression and an Attrition-Based model, supporting the HR department in anticipating staffing requirements and improving recruitment planning.

Metric	Value
Total Employees Analysed	311
Active Employees	207 (66.6%)
Terminated Employees	104 (33.4%)
Departments	6
Hire Year Range	2006 – 2018
Peak Hiring Year	2011 (83 hires)
Org-wide Replacement Hires/Year	~66 (attrition model)
Overall Model R^2	0.674
LOO-CV Mean MAE	16.76 hires

2. Objective

The objective of this project is to forecast future hiring needs by analysing historical recruitment data. By identifying patterns in past hiring activity, attrition behaviour, and departmental trends, this analysis helps the HR department to:

- Anticipate staffing requirements before vacancies arise
- Allocate recruitment budgets more effectively across departments
- Identify departments at highest risk of workforce shortfall
- Support strategic decision-making around talent acquisition
- Reduce time-to-hire by preparing recruitment pipelines in advance

3. Data Source

The dataset used in this project is the publicly available Human Resources Data Set from Kaggle, created by Dr. Carla Patalano and Rich Huebner for educational purposes.

Item	Detail
Dataset Name	HRDataset_v14.csv
Source	Kaggle — Human Resources Data Set
Original Records	311 employees, 36 columns
Final Columns (after cleaning)	41 (36 original + 5 derived)
Hire Year Range	2006 – 2018
Departments	Production, IT/IS, Sales, Software Engineering, Admin Offices, Executive Office
Key Columns Used	DateofHire, DateofTermination, Department, RecruitmentSource, TermReason, TenureYears

4. Methodology

The analysis followed a structured four-step pipeline executed across four Jupyter notebooks.

4.1 Step 1 — Data Cleaning (Notebook 01)

The raw dataset was cleaned and standardised before analysis.

Key tasks included:

- Parsed DateofHire and DateofTermination from strings to datetime objects
- Handled 207 missing DateofTermination values — these are active employees, not data gaps
- Investigated and filled 8 missing ManagerIDs — confirmed as Webster Butler's ID (39)
- Stripped whitespace and applied Title Case to 10 text columns
- Manually fixed acronyms broken by .title() — IT/IS and Still Employed
- Converted identifier columns (EmpID, ManagerID, etc.) from int to string

- Added 5 derived columns: HireYear, HireMonth, HireMonthName, TenureYears, IsActive

4.2 Step 2 — Exploratory Data Analysis (Notebook 02)

Nine visualisations were produced to explore hiring trends, seasonal patterns, attrition behaviour, and recruitment channel effectiveness.

4.3 Step 3 — Forecasting Model (Notebook 03)

Two complementary forecasting models were built and evaluated:

- **Model 1** — Linear Regression: trained on 2011–2016 annual hire counts to project overall hiring direction. The training window was selected because 2006–2010 had very few records and 2017–2018 showed data tail-off.
- **Model 2** — Attrition-Based Model: estimates annual replacement hires per department using Active Headcount x Historical Attrition Rate. This is more actionable for department-level planning.
- Model accuracy evaluated using R2, MAE, RMSE, MAPE, and Leave-One-Out Cross Validation.

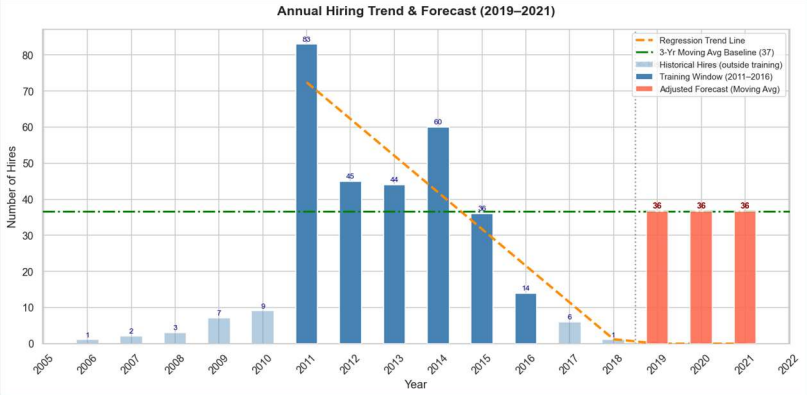
4.4 Step 4 — Report (Notebook 04)

All findings were consolidated into this report with visualisations, model results, and actionable HR recommendations.

5. Key Findings

5.1 Annual Hiring Trend

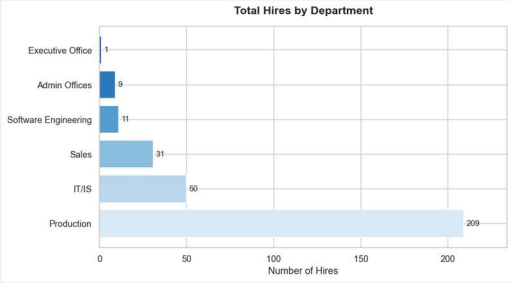
Hiring activity peaked sharply in 2011 with 83 hires, likely reflecting a major organisational expansion phase. It declined steadily to 14 hires in 2016 and just 1 hire in 2018. The linear regression slope of -10.17 hires per year confirms a strong downward trend. Since the raw regression forecast goes negative by 2019, the 3-year moving average of 37 hires per year is used as the adjusted planning baseline.

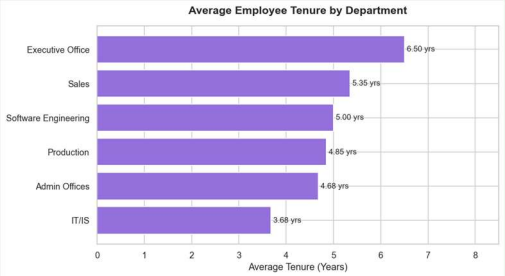


Annual Hiring Trend with Regression Line and 3-Year Moving Average Forecast]

5.2 Departmental Analysis

Production is by far the largest department with 209 total hires representing 67% of all hiring activity. It also has the highest attrition rate at 39.7%, meaning nearly 4 in 10 Production employees have left the organisation — creating the greatest ongoing replacement demand. Software Engineering has the second-highest attrition at 36.4% despite being a small department of only 11 total hires. IT/IS has the shortest average tenure of 3.68 years, indicating the fastest employee churn cycle.



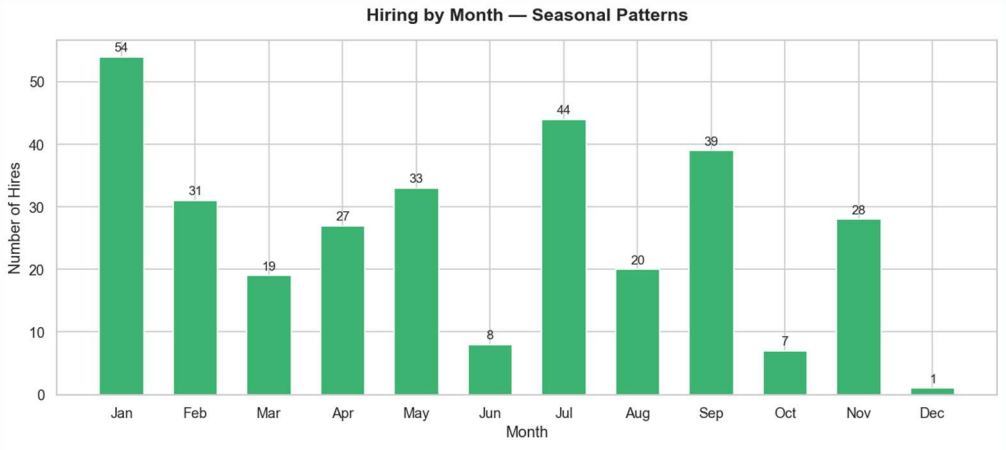


Total Hires by Department (left) and Attrition Rate by Department (right)

Department	Total Hires	Active	Terminated	Attrition Rate	Avg Tenure (yrs)
Production	209	126	83	39.7%	4.85
IT/IS	50	40	10	20.0%	3.68
Sales	31	26	5	16.1%	5.35
Software Engineering	11	7	4	36.4%	5.00
Admin Offices	9	7	2	22.2%	4.68
Executive Office	1	1	0	0.0%	6.50

5.3 Seasonal Hiring Patterns

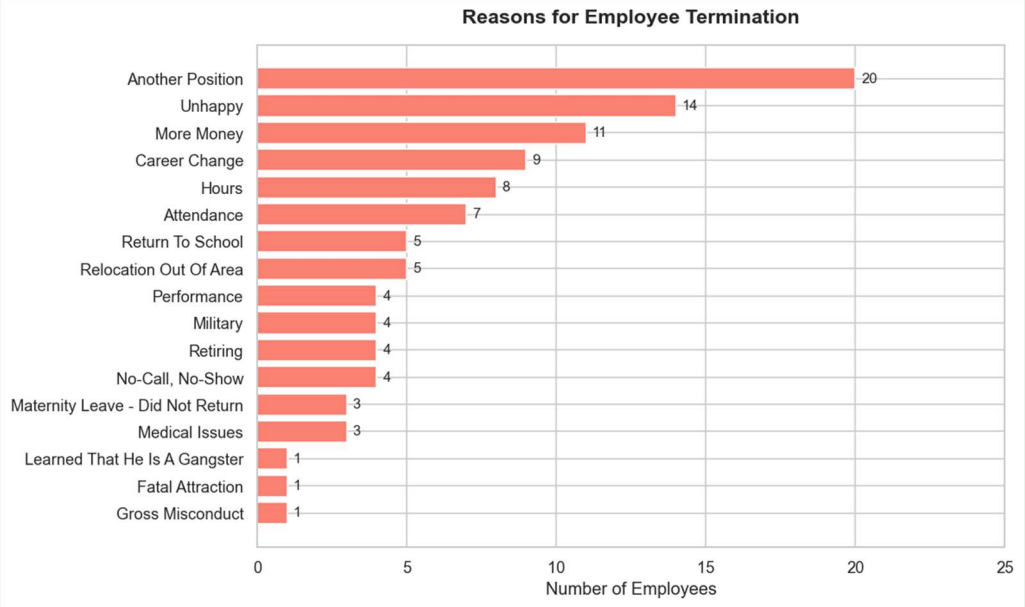
Clear seasonal hiring patterns exist in the data. January is the strongest hiring month with 54 hires, followed by July (44) and September (39). December (1), October (7), and June (8) are consistently the quietest months. This indicates HR should prepare recruitment pipelines 4-6 weeks before the January, July, and September peaks.



Monthly Hiring Patterns — Peak and Quiet Months

5.4 Termination Reasons

The majority of employee exits are voluntary. The top three reasons are Another Position (20 employees), Unhappy (14 employees), and More Money (11 employees) — together accounting for 43% of all terminations. This suggests a significant proportion of attrition is preventable through targeted retention strategies addressing compensation, job satisfaction, and career development.

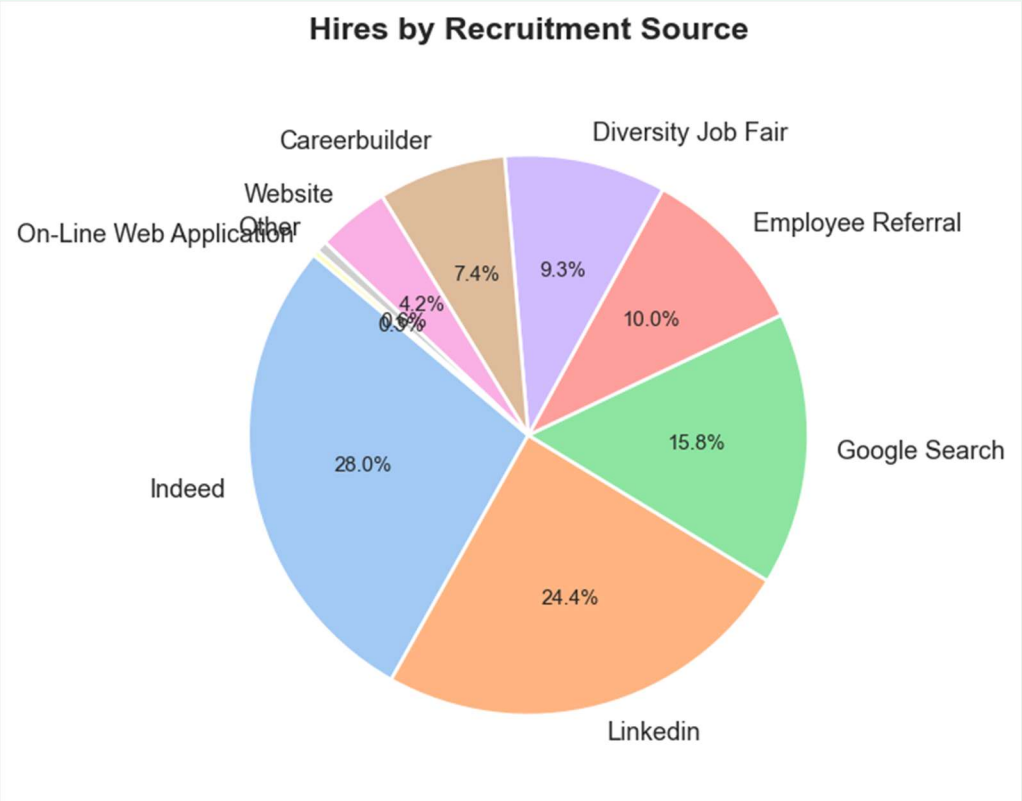


Employee Termination Reasons

Termination Reason	Count	% of Terminated
Another Position	20	19.2%
Unhappy	14	13.5%
More Money	11	10.6%
Career Change	9	8.7%
Hours	8	7.7%
Attendance	7	6.7%
Other reasons	35	33.7%

5.5 Recruitment Source Effectiveness

Indeed (87 hires, 28.0%) and LinkedIn (76 hires, 24.4%) are the two dominant recruitment channels, together accounting for over 52% of all hires. Google Search (49), Employee Referral (31), and Diversity Job Fair (29) are the next most effective channels. HR budget allocation for job postings should reflect this distribution.



Recruitment Sources — Volume (left) and Share (right)

Recruitment Source	Hires	Percentage
Indeed	87	28.0%
LinkedIn	76	24.4%
Google Search	49	15.8%
Employee Referral	31	10.0%
Diversity Job Fair	29	9.3%

Recruitment Source	Hires	Percentage
CareerBuilder	23	7.4%
Website	13	4.2%
Other / Web App	3	1.0%

6. Forecasting Model Results

6.1 Model 1 — Linear Regression

A Linear Regression model was trained on annual hire counts from 2011 to 2016 (6 data points). The model identified a clear negative trend with a slope of -10.17 hires per year.

Parameter	Value
Training Period	2011 – 2016
Slope (trend per year)	-10.17 hires
Intercept	20,527.17
R^2 Score	0.674
MAE	10.97 hires
RMSE	12.09 hires
MAPE	27.6%
LOO-CV Mean MAE	16.76 hires
Raw Forecast 2019	-8.9 (clipped to 0 — trend went negative)
Adjusted Baseline	37 hires/year (3-year moving average)

Note: The raw regression forecast produces negative values by 2019 due to the sharp hiring drop between 2011 and 2016. The 3-year moving average of 37 hires per year is used as the adjusted planning baseline instead.

6.2 Model 2 — Attrition-Based Forecast

The attrition-based model estimates annual replacement hiring needs per department using: Predicted Hires = Active Headcount x Attrition Rate. This model is more actionable for department-level HR planning.

Department	Active Staff	Attrition Rate	Predicted Hires/Year	Priority
Production	126	39.7%	50.0	HIGH
IT/IS	40	20.0%	8.0	MEDIUM
Sales	26	16.1%	4.2	LOW
Software Engineering	7	36.4%	2.5	HIGH
Admin Offices	7	22.2%	1.6	MEDIUM
Executive Office	1	0.0%	0.0	LOW
TOTAL	**207**	**—**	**66.3**	**—**

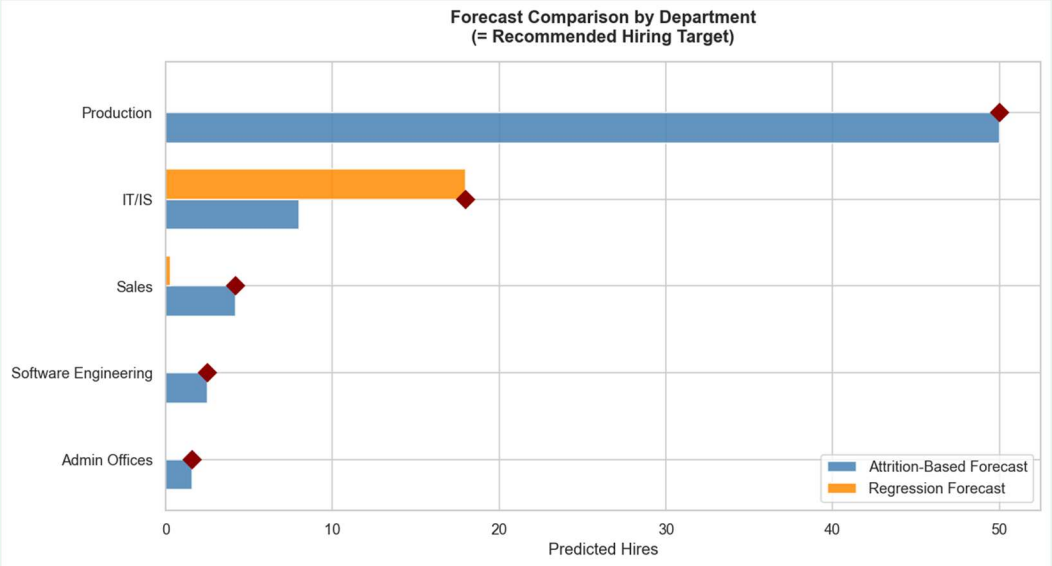
6.3 Model Accuracy Evaluation

Model accuracy was evaluated using four standard regression metrics and Leave-One-Out Cross Validation — the recommended technique for very small datasets.

Metric	Overall LR	Production	IT/IS	Sales	Software Engg
R^2	0.674	0.826	0.280	0.180	0.812
MAE	10.97	6.80	4.33	2.19	0.56
RMSE	12.09	8.45	5.48	3.02	0.63
MAPE	27.6%	26.3%	129.4%	67.8%	31.0%
LOO-CV MAE	16.76	—	—	—	—

Production and Software Engineering show strong R^2 scores above 0.80 — the linear trend fits well for these departments. IT/IS and Sales show weak R^2 below 0.30, reflecting irregular hiring patterns — the attrition-based model is preferred for these departments.

6.4 Combined Forecast Summary



Model Comparison and Recommended Hiring Targets by Department

7. Recommendations

Based on the analysis and forecasting results, the following recommendations are made for HR workforce planning:

Priority	Department / Area	Recommendation
HIGH	Production	Pre-emptively hire ~50 employees/year to offset 39.7% attrition. Build a continuous recruitment pipeline — do not wait for vacancies to arise.
HIGH	Software Engineering	Despite being a small team, 36.4% attrition requires ~2-3 hires/year. Prioritise retention through career development and technical mentorship.
MEDIUM	IT/IS	IT/IS shows a growing regression trend (slope +2.0). Plan for 8-18 hires/year. Focus sourcing on LinkedIn and Indeed for technical talent.
MEDIUM	Admin Offices	Attrition at 22.2% — plan for ~2 replacement hires/year. Small team so each departure has high operational impact.

Priority	Department / Area	Recommendation
MEDIUM	Seasonal Planning	January, July, and September are peak hiring months. Launch recruitment campaigns in December, June, and August to build pipeline ahead of peaks.
LOW	Sales	Most stable department (16.1% attrition, 5.35-year avg tenure). Maintain current practices. Plan for ~4 replacement hires/year.
ALL	Retention Strategy	"Another Position" and "Unhappy" are top exit reasons. Introduce stay interviews, career pathing, and compensation reviews to reduce voluntary attrition.
ALL	Recruitment Budget	Indeed (28%) and LinkedIn (24.4%) drive over 52% of hires. Concentrate job posting budget on these two platforms for maximum return.

8. Conclusion

This project successfully demonstrated a complete predictive workforce planning pipeline using historical HR recruitment data. The analysis progressed through four structured stages — data cleaning, exploratory analysis, forecasting modelling, and reporting — producing actionable insights at both the organisational and departmental level.

Five key conclusions from the analysis:

- Hiring has declined significantly since the 2011 peak. The 3-year moving average of 37 hires per year serves as the most realistic near-term planning baseline.
- Production drives workforce demand — 67% of all historical hires and a 39.7% attrition rate mean this department requires a continuous, proactive recruitment strategy with approximately 50 replacement hires per year.
- Voluntary attrition is largely preventable — 43% of terminations cite "Another Position", "Unhappy", or "More Money". Targeted retention programmes could meaningfully reduce future hiring costs.
- Seasonal planning matters — January, July, and September consistently show peak hiring activity. Recruitment campaigns should be launched 4-6 weeks before these peaks.

- The attrition-based model is the more reliable forecasting tool for department-level planning, predicting approximately 66 total replacement hires per year organisation-wide.

9. Limitations

- The training dataset contains only 6 years of complete data (2011-2016), which limits the statistical power of the regression model.
- The linear regression model cannot capture non-linear hiring patterns seen in IT/IS and Sales departments.
- The attrition-based model uses historical attrition rates as a proxy for future turnover — external factors such as economic conditions or company strategy changes are not accounted for.
- The dataset represents a single fictional organisation — findings should not be generalised without validation on real-world data.

10. Deliverables

Deliverable	File	Description
Cleaned Dataset	data/processed/HRDataset_Cleaned.csv	311 records, 41 columns
Data Cleaning	notebooks/01_data_cleaning.ipynb	Full cleaning pipeline
EDA	notebooks/02_exploratory_analysis.ipynb	9 analysis charts
Forecasting Model	notebooks/03_forecasting_model.ipynb	LR + Attrition models
Final Report NB	notebooks/04_vizualization.ipynb	Report notebook
Project Report	Workforce_Planning_Report.pdf	This document
Charts (7 files)	outputs/figures/	PNG visualisations
README	README.md	Project documentation

End of Report